What will the Type 31 frigates cost to own?

Whole-life cost, sensitivity and risk analysis from public data

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Illustrative independent analysis. All costs £m in approximate 2023-24 prices unless stated.

THE ANSWER

The five ships will cost about £6.0bn over their lives – and buying them is only 38% of that

£6.0bn
Whole-life cost, point estimate (undiscounted)

62%
Share spent on running and supporting the ships

£6.7bn
P80: 80% chance the cost is at or below this

£698m
Contingency needed above the point estimate for P80

- **Acquisition** of £2.3bn (the £2.2bn approved budget plus 5% growth) is the visible part of the bill. **Support** – crew, fuel, maintenance, in-service support, a mid-life update and disposal – adds £3.7bn over 25 years per ship.
- Using only the direct costs the UK publishes gives £4.8bn. That would understate support compared with every international comparator, so the model adds indirect support calibrated to the US Navy's FFG-62 frigate.
- Allowing for uncertainty, the median outcome is £6.3bn, and there is a 72% chance the final cost is higher than the point estimate – most risks push costs up, not down.

Source: Type31_WholeLifeCost_Risk_Model.xlsx (Summary, Risk_Results). Monte Carlo figures are from one 2,000-iteration run.

APPROACH

A transparent model: published UK data, calibrated to overseas programmes, then stress-tested

1 Inputs

Every number in one sheet, tagged Sourced, Derived, Benchmark or Assumption, with low / base / high values.

2 Cost model

Year-by-year profile 2019-2065 for each ship: build spend, crew and fuel, maintenance, indirect support, mid-life update, disposal.

3 Benchmarking

Support share and running cost checked against US (GAO, FFG-62), Canadian (CSC) and Australian (ANZAC) evidence.

4 Sensitivity

Each uncertain input moved to its low and high value to rank what drives the answer.

5 Risk

2,000 Monte Carlo simulations using triangular distributions to give a confidence range (P10-P90).

Core equation: WLC = Acquisition × (1 + growth) + Ships × Life × (Operating + Maintenance) × (1 + Indirect uplift) + Ships × (Mid-life update + Disposal)

Present value: future costs discounted at the Green Book 3.5% real rate give £3.1bn from 2026 onward. A built-in check confirms the year-by-year total equals the closed-form equation.

Main data sources

- UK MoD letter to Public Accounts Committee (2023): £2.2bn budget, 5 ships
- MoD FOI 2015/06440: Type 23 direct operating cost
- Written PQ (2025): Type 23 upkeep spend since 2014
- Royal Navy / GOV.UK: Type 23 and Type 31 crew sizes
- US DoD FFG-62 Selected Acquisition Report (2022)
- US GAO-20-2; Canadian PBO (2022); ANAO (2019)

Sources listed in the workbook (Sources sheet): MoD, HM Treasury, Royal Navy, US DoD, US GAO, Canadian PBO, ANAO.

KEY ASSUMPTIONS

Every assumption is explicit, evidenced and rated – the least certain ones are tested hardest

Input	Base (range)	Basis	Confidence
Acquisition budget	£2.2bn	MoD letter to Public Accounts Committee, May 2023	High
Build cost growth	5% (0-20%)	Inflation since 2022 not in budget; contractor reported losses	Medium
Service life	25 yrs (20-30)	Standard frigate planning life; matches US FFG-62	Medium
Crew: Type 23 → Type 31	185 → 105 (100-115)	Royal Navy ship pages; GOV.UK	High
Running cost (direct)	£10.7m / ship-yr	Type 23 FOI cost (£11.7m, 2013-14) inflated and scaled for crew	Medium
Maintenance	£5.7m / ship-yr	Type 23 upkeep spend £941m since 2014, scaled to 80%	Low
Indirect support uplift	0.62 (0.30-0.82)	US FFG-62 cost split: sustaining + other ÷ direct	Medium
Mid-life update	£75m (£50-150m)	Type 23 refits (HMS Iron Duke > £100m)	Low
Discount rate	3.5% real	HM Treasury Green Book	High

Where the Type 23 crew of 185 comes from

The Royal Navy's page for HMS St Albans states the ship is sailed by a crew of 185, with accommodation for up to 205. Crew sizes vary between ships and over time, so 185 is used as the class figure. A 10% error in it would move the total by under 2%.

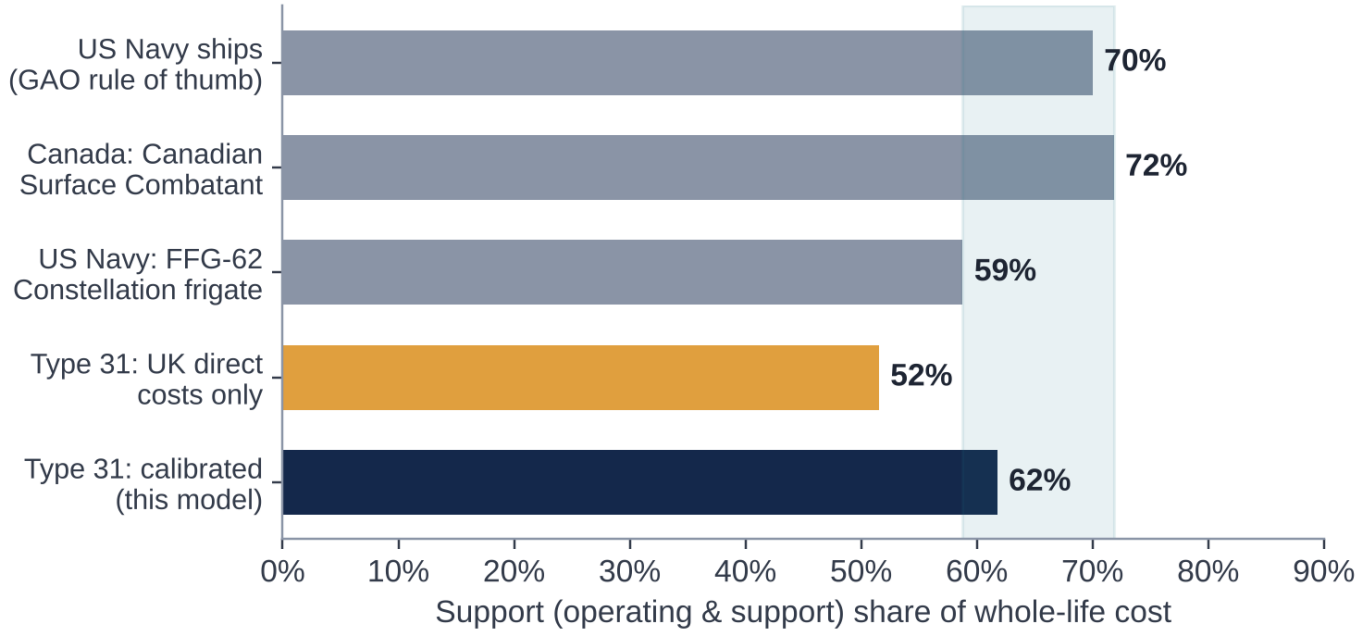
Why the crew ratio matters

Type 31 is designed for about 105 sailors (GOV.UK), with the Royal Navy now citing around 115. Scaling Type 23 crew costs by 105/185 is the main reason a Type 31 is estimated to be cheaper to run than the ship it replaces.

Full reasoning on the Assumptions_Log sheet. Type 23 crew of 185: Royal Navy, HMS St Albans ship page ('a crew of 185, but with accommodation for up to 205').

INTERNATIONAL BENCHMARKS

UK direct costs alone would understate support; calibrating to overseas frigates brings the estimate into line



Why calibrate?

The MoD's published running costs exclude base support, training, spares and overheads. On direct costs alone support is 52% of whole-life cost – below every comparator.

How

In the US Navy's FFG-62 estimate, sustaining support and other indirect costs add 62% on top of crew, operations and maintenance. Applying that ratio lifts the Type 31 support share to 62%.

Cross-check

Calibrated Type 31 running costs equal 5.8% of unit build cost per year, almost identical to FFG-62's 5.7%.

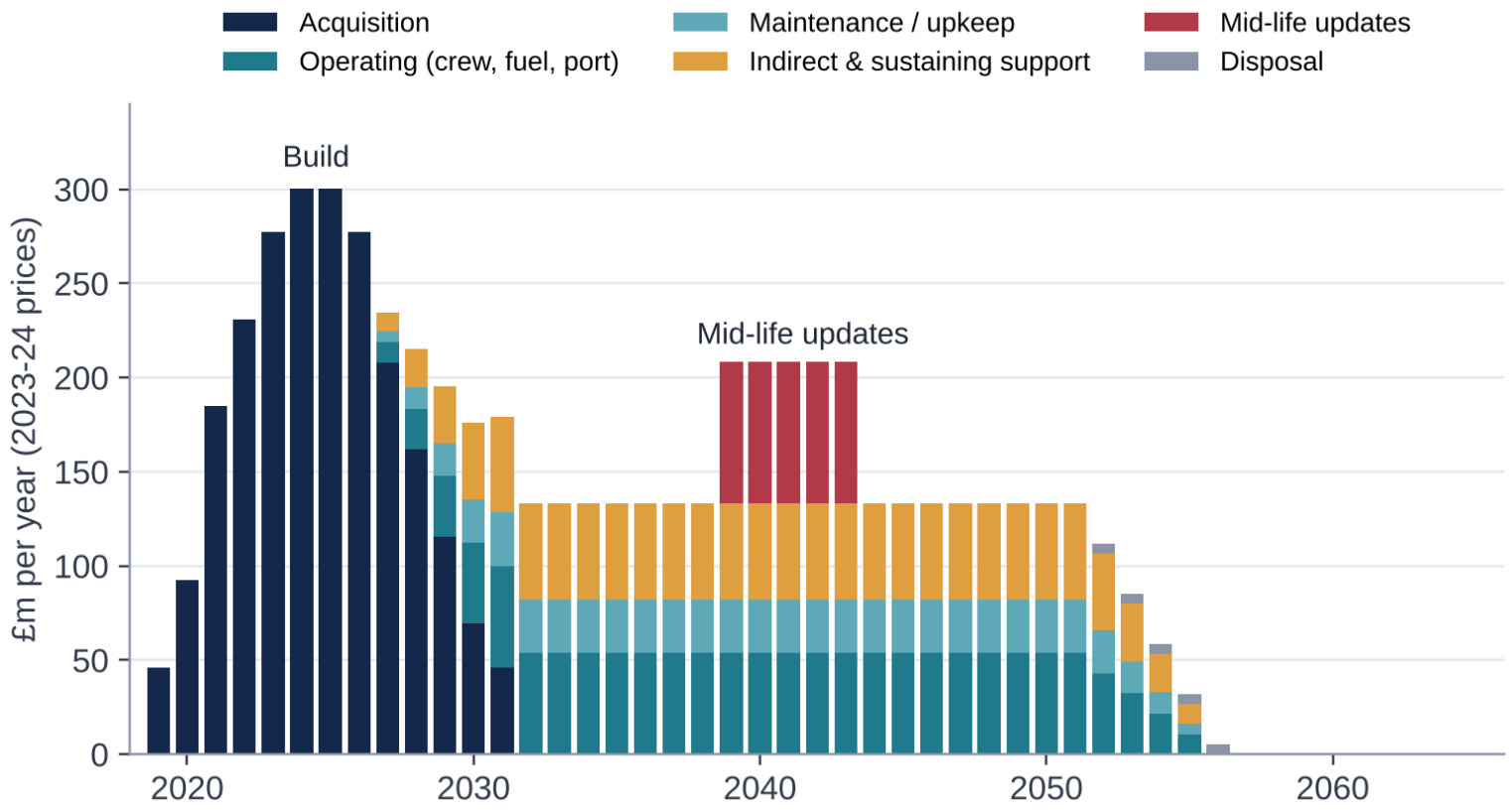
Context

Australia's eight ANZAC frigates had a 2018-19 sustainment budget of A$374m – about A$47m per ship per year – showing support runs at tens of millions per ship.

Sources: GAO-20-2 (2019); Canadian PBO, Life Cycle Cost of the Canadian Surface Combatants (2022); FFG(62) Selected Acquisition Report (Dec 2022); ANAO Anzac sustainment (2019).

COST PROFILE

Running costs dominate: support is 1.6× the purchase price over the ships' lives



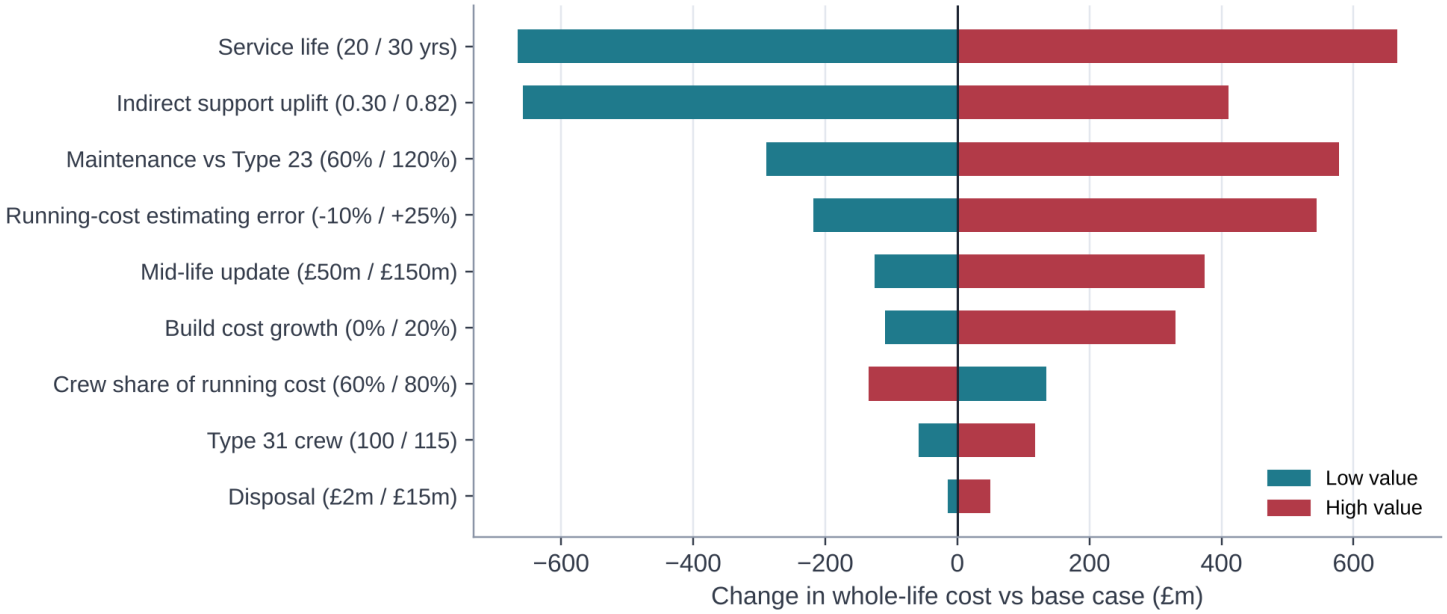
Category	£m	Share
Acquisition	2,310	38%
Operating (crew, fuel, port)	1,343	22%
Maintenance / upkeep	713	12%
Indirect & sustaining support	1,274	21%
Mid-life updates	375	6%
Disposal	25	0%
Total	**6,040**	**100%**

Per ship: **£1,208m** over its life, against a build cost of about £460m.

Source: WLC_Model sheet. Ships enter service 2027-2031 and leave 2052-2056; mid-life updates in each ship's 13th year.

SENSITIVITY

Service life and support assumptions move the answer far more than the build contract



Biggest driver

Service life: 20 vs 30 years swings the total by £1,332m. Longer life costs more in total but spreads the build cost over more years of use.

Next

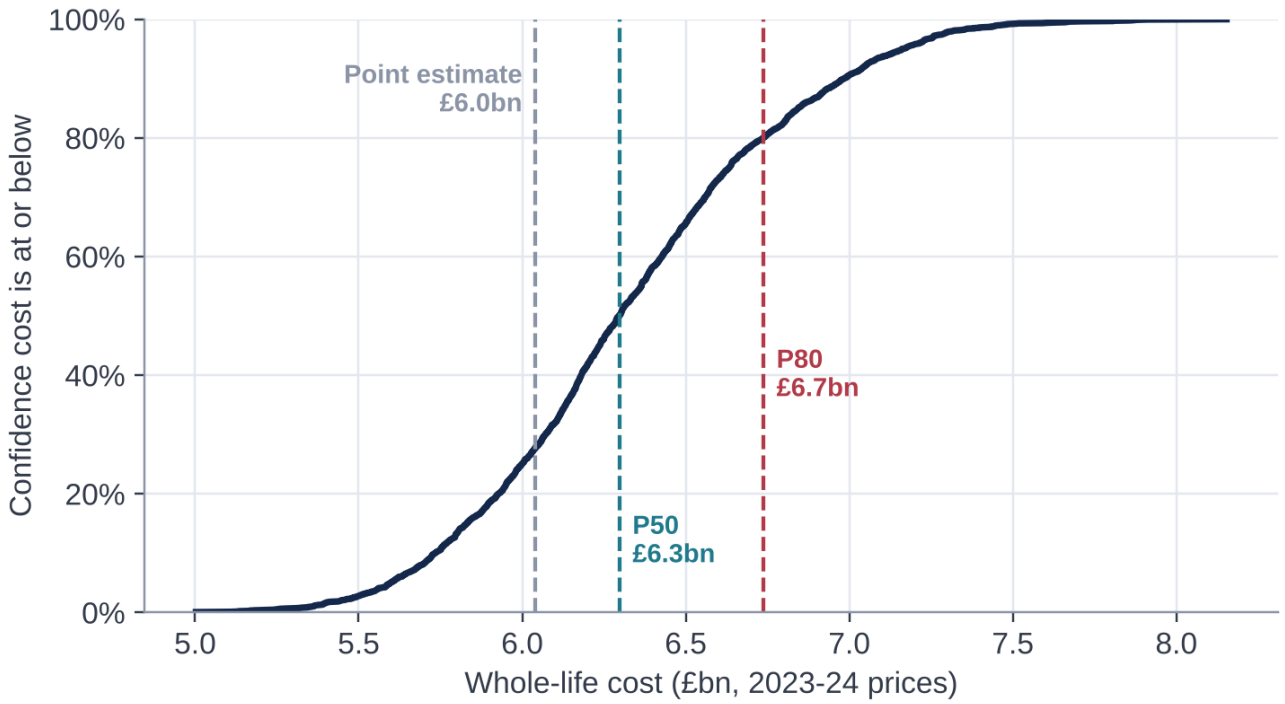
Indirect support and maintenance assumptions each swing the total by about £1.1bn or less – the least-evidenced inputs, so the most valuable to firm up.

Least important

Crew size, crew share of cost and disposal matter little: the Type 31's small crew is already locked into the design.

Source: Sensitivity sheet. Each input moved to its low and high value with all others held at base.

Budget at P80 (£6.7bn) and focus scrutiny on support costs



P10 £5.7bn · **P50** £6.3bn · **P80** £6.7bn · **P90** £7.0bn

1. Hold contingency

Plan to P80, about £698m (12%) above the point estimate. 72% of simulated outcomes exceed the point estimate.

2. Publish the support case

The through-life support business case is the biggest gap in public data; it would replace the least certain assumptions.

3. Decide life early

Service life is the largest driver. Planning for 25 vs 30 years changes crewing, upkeep and replacement timing.

4. Protect the small crew

Low crewing is a key saving versus the Type 23 – each extra 10 sailors adds roughly £0.1bn over the class's life.

Limitations: direct UK running costs are estimated by analogy with the Type 23; overseas cost structures differ from the UK's; the £2.2bn budget mixes price years.